ACL Seminar 3: Zipf's Law Across Literary & Academic Corpora

Comparative Study: Mary Shelley, Carlo Collodi, Brown Corpus, and UMFST Bachelor's Thesis

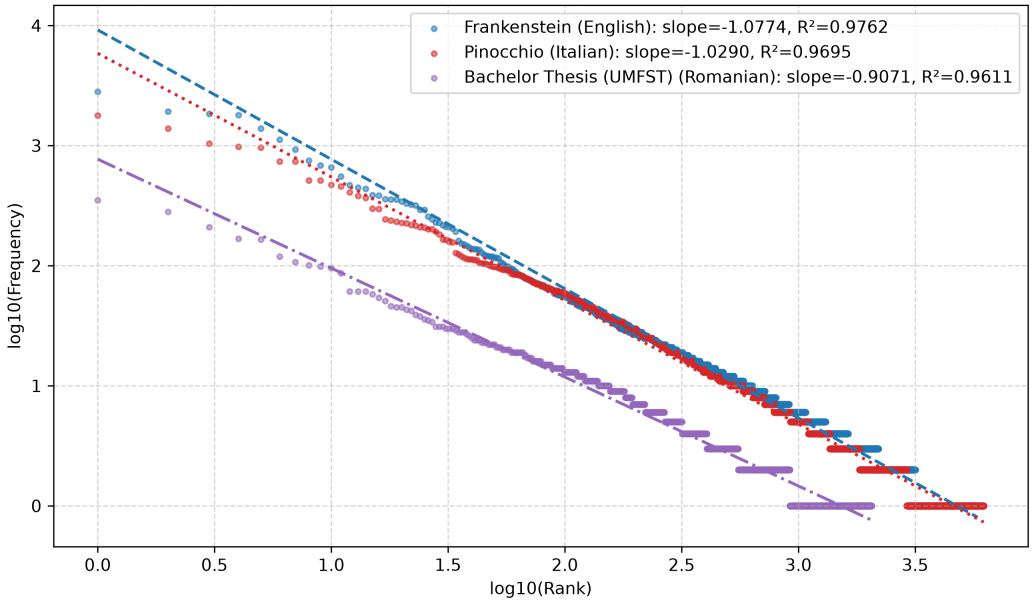
1. Mathematical Formulation

Zipf's Law: f(r) = C / r^s. Log-transformed: log10(f) = intercept + slope * log10(r).

2. Experimental Results

Corpus	Lang	Tokens	Types	TTR	Top Word	Slope	s	R^2
Frankenstein	English	50,000	5,849	0.1170	the (2,821)	-1.0774	1.0774	0.9762
Pinocchio	Italian	41,525	6,215	0.1497	e (1,788)	-1.0290	1.0290	0.9695
Brown Corpus	America	50,000	8,246	0.1649	the (3,633)	-0.9795	0.9795	0.9693
Bachelor Thesis (UMFST)	Romania	8,301	2,056	0.2477	de (351)	-0.9071	0.9071	0.9611

Graph 1: Zipf's Law – Cross-Linguistic Comparison (English, Italian, Romanian)



3. Conclusions

Zipf's law holds across all four corpora (R^2 >= 0.961). The Romanian technical Bachelor's thesis exhibits the highest lexical diversity (TTR = 0.2477) due to specialized terminology.